

Cybersecurity and Data Protection

1. Overview

The system is delivered with cybersecurity controls covering data integrity, prevention of unauthorised access, secured communications and ongoing threat management.

The unit is fully self-contained for connectivity. It sends data to the secure cloud service over the internet using its own internal communications hardware. It does not connect to, and does not require any access to, the client's WiFi network. This keeps the detection system entirely separate from the client's internal network.

2. How the System Connects

The unit initiates outbound connections only; it exposes no incoming access points to the internet.

All data sent from the unit to the cloud service is encrypted in transit.

Technical support access is made by the unit reaching out to the secure cloud service

3. Access Control and Unit Hardening

Control	Protection provided
Login security	Management access uses cryptographic keys only; passwords and administrator logins are disabled on the unit.
Firewall	Blocks all unsolicited incoming connections by default.
Brute-force defence	Automatically blocks sources that attempt repeated failed logins.
Automatic updates	Security updates are applied automatically.
Reduced attack surface	Unused features and interfaces are switched off.
Wireless encryption	The unit's internal wireless link uses WPA2 with AES encryption.

4. Cloud Dashboard Security

Control	Protection provided
Encrypted access	The dashboard is served over HTTPS only, with automatically managed certificates.
Password protection	User passwords are stored only as secure one-way hashes, never in plain text.
Session security	Encrypted, time-limited sessions (8 hours) that require re-login; protected against common session
Form protection	All changes are protected against cross-site request forgery.
Login lockout	Accounts lock after repeated failed attempts from an address.
Role-based access	Separate Administrator, Operator and Viewer roles limit what each user can do.
Verified data only	Data sent to the dashboard is authenticated with a shared secret, so only the lightning detection unit can submit records.

5. Data Protection

Records are stored on the unit first, so the on-site record is complete even during an internet outage, then confirmed to the secure cloud service over an encrypted connection.

Interference from nearby electronics is filtered out so that only genuine lightning is recorded and alerted, protecting the accuracy of the record and avoiding false notifications.

Only the minimum personal data is held: recipient names and phone numbers for SMS notifications. No other personal data is collected, and access is restricted by role.

6. Threats and Mitigations

Threat	Mitigation
Unauthorised remote access	Key-only access, no passwords, no incoming ports on site, no use of client network.
Password guessing	Secure hashing, account lockout, encrypted sessions.
Interception of data	Encryption on all internet and wireless links.
Fake or spoofed data	Authenticated data submission with a shared secret.
Known software vulnerabilities	Automatic security updates.
Tampering with the unit	Outbound-only design; access keys can be revoked centrally; a storage swap restores a clean, trusted configuration.